

Translated

From: e-maxx.ru

Dijkstra on sparse graphs

For the statement of the problem, the algorithm with implementation and proof can be found on the article [Dijkstra's algorithm](#).

Algorithm

We recall in the derivation of the complexity of Dijkstra's algorithm we used two factors: the time of finding the unmarked vertex with the smallest distance $d[v]$, and the time of the relaxation, i.e. the time of changing the values $d[\text{to}]$.

In the simplest implementation these operations require $O(n)$ and $O(1)$ time. Therefore, since we perform the first operation $O(n)$ times, and the second one $O(m)$ times, we obtained the complexity $O(n^2 + m)$.

It is clear, that this complexity is optimal for a dense graph, i.e. when $m \approx n^2$. However in sparse graphs, when m is much smaller than the maximal number of edges n^2 , the complexity gets less optimal because of the first term. Thus it is necessary to improve the execution time of the first operation (and of course without greatly affecting the second operation by much).

To accomplish that we can use a variation of multiple auxiliary data structures. The most efficient is the **Fibonacci heap**, which allows the first operation to run in $O(\log n)$, and the second operation in $O(1)$. Therefore we will get the complexity $O(n \log n + m)$ for Dijkstra's algorithm, which is also the theoretical minimum for the shortest path search problem. Therefore this algorithm works optimal, and Fibonacci heaps are the optimal data structure. There doesn't exist any data structure, that can perform both operations in $O(1)$, because this would also allow to sort a list of random numbers in linear time, which is impossible. Interestingly there exists an algorithm by Thorup that finds the shortest path in $O(m)$ time, however only works for integer weights, and uses a completely different idea. So this doesn't lead to any contradictions. Fibonacci heaps provide the optimal complexity for this task. However they are quite complex to implement, and also have a quite large hidden constant.